

## Project Proposal



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# Thesis Evaluation System

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## **DECLARATION**

The project proposal is submitted to the Department of Information and Communication Engineering, Noakhali Science and Technology University. So, I declare that this project report has not been submitted elsewhere for the requirement of any kind of degree, diploma and publication.

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## ACCEPTANCE

The project proposal is submitted to the Department of Information and Communication Engineering, Noakhali Science and Technology University, Sonapur, Noakhali.

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## ABSTRACT

The web-based thesis evaluation system is designed to automate and streamline the thesis submission and evaluation process in academic institutions. Traditional thesis evaluation methods often rely on manual paperwork, email communication and physical record keeping, which can lead to delays, miscommunication, data loss and lack of transparency. To address these issues, the proposed system provides a centralized digital platform for managing the entire thesis evaluation workflow. The system allows students to submit their theses online and track evaluation status. Supervisors can review submissions, provide feedback and approve or recommend revisions. Evaluators can assess assigned theses and submit marks and comments digitally. Administrators can manage users, assign roles, publish results and generate reports. The system ensures role-based access control, secure data storage and efficient record management. By implementing this project, we will be able to reduce administrative workload, improve transparency, minimize errors and ensure timely evaluation and result publication.

**Keywords:** Thesis Evaluation, Automated System, Thesis Report Generation

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## Abbreviations

HTML-Hyper Text Markup Language  
CSS-Cascading Style Sheets  
ER Diagram-Entity Relationship Diagram  
UI- User Interface  
JS-Javascript  
SSPL-Server-Side Public License

# **CHAPTER 1**

## **INTRODUCTION**

### **1.1 Introduction**

The evaluation of thesis is an important process in the academic institutions today which evaluates the research work and performance of the students. Conventionally, the process is done manually where paperwork, emails and even in-person communication is involved and consequently results to delays, mismanagement of data, lack of transparency and the addition of administrative workload. In order to address these constraints, a Thesis Evaluation Web Portal is suggested as an effective and dependable online tool.

The thesis evaluation Web Portal is an online tool that manages the process of submitting, supervising, assessing and approving student theses. It offers a central database on which students may post their thesis paper, supervisors may view and comment on those papers and evaluators can review and mark the thesis on the internet. The system also guarantees role-based accessibility which provides every user, including students, supervisors, evaluators and administrators, the opportunity to conduct certain roles safely.

The portal enhances transparency, lowers processing time, and human errors by fully automating the thesis evaluation process. It is also able to keep good records of submissions, evaluation and outcomes making the process to be much more structured and readily available. On the whole, the Thesis Evaluation Web Portal can improve the efficiency of academic activities and promote the use of the paperless evaluation system in the educational institutions of the present day.

### **1.2 Background Information**

Thesis evaluation is an important academic activity in institutions of higher learning, which are applied to identify the ability of the students to conduct research, possess analytical skills and knowledge on particular subjects. Conventionally, it has been a manual process that requires paper-based submission, physical record maintaining, email and face-to-face coordinating systems between the students, supervisors and evaluators. Although the aforementioned approach has been employed over a number of years, there are a number of challenges that it poses in the current academic landscape that is rapidly expanding.

Thesis evaluation systems that are done manually are time consuming and subject to inefficiency. Students can experience a slow feedback and publication of results, and supervisors and evaluators have to work with huge amounts of documents by hand. Also, it becomes hard to monitor thesis status, version control, data security and academic records when managed in a non-centralized digital environment. These problems may cause miscommunication, loss of data, absence of transparency and more administrative effort.

### 1.3 Problem Statement

In many academic institutions, the thesis evaluation process is still conducted using manual or semi-digital methods such as paper-based submissions, email communication and disconnected systems. These traditional approaches often result in delays, miscommunication, data redundancy and difficulty in tracking the evaluation progress. Students face uncertainty regarding submission status and feedback, while supervisors and evaluators experience challenges in managing multiple theses efficiently.

The absence of a centralized and automated system makes it difficult to maintain transparency, ensure secure data storage, and generate structured evaluation reports. Manual handling also increases the risk of data loss, human error and inconsistency in grading.

Therefore, there is a need to develop a **Thesis Evaluation Web Portal** that provides a secure, centralized, and role-based platform to manage thesis submission, supervision, evaluation and result publication efficiently. The proposed system aims to streamline the entire workflow, reduce administrative burden and improve transparency and accuracy in the thesis evaluation process.

### 1.4 Motivation of the Work

The lack of efficiency and the difficulties of the thesis submission that requires manual preparations, stimulates the creation of an automated solution. Not only can the time and efforts be saved using an automation tool but the overall satisfaction of teachers and students can also be increased due to the fair and systematic organization of this.

In this project, the following considerations drive the project:

**1.Administrative Burden:** The process of automation of thesis submission will release the large administrative time and resources.

**2.Enhancing Fairness:** The system makes sure that there is equity by taking into consideration teacher priorities and availability.

**3.Enhancement of Flexibility:** The system is flexible and can be altered rapidly should there be changes or emergencies.

**4.Transparency:** The thesis accessible to teachers and students in real-time, which will help to minimize confusion.

## **1.5 Project Objectives**

The primary goal of the Thesis Evaluation Web Portal is to create and develop a web-based system, which will be centralized and that will automate and simplify the thesis filing and assessment process in higher learning institutions. The following are the specific goals of the system:

1. To enable students to submit their thesis documents over the internet platform.
2. To enable the assessors to analyze theses and submit evaluation report and marks electronically.
3. To minimize paper work and administrative overhead by automating.
4. To have proper and safe records of thesis submissions, assessments and outcomes.
5. To provide role-based access control to protect the data security and integrity.
6. In order to produce reports and evaluation summaries to be used by academics and administration.

## **1.6 Project Scope**

The functional boundaries and capabilities of the proposed system are determined by the scope of Thesis Evaluation Web Portal. This project aims at creating a centralized and web-based platform which will automatically process and control the thesis submission and assessment in a university.

The system will enable the students to enroll and log in to the system and place their thesis documents online beside monitoring the progress of their evaluation and accessing feedback and results. The supervisors will be given opportunity to check the submitted theses, comment or approve changes and monitor the progress of the students. The system will provide evaluators with the access to the assigned theses and allow submitting evaluation reports and marks. The administrators will handle user accounts, appoint supervisors and evaluators and restrict the settings of the system as well as produce academic reports.

## **CHAPTER 2**

### **LITERATURE REVIEW**

#### **2.1 Introduction**

As the information technology is expanding fast and the number of the students rises, it is anticipated that a digital solution capable of making the thesis evaluation process easier and automated is highly demanded. The Thesis Evaluation Web Portal is an online platform that aims at handling thesis submission, supervision, assessment and publication of results in an effective and well-organized way.

The portal will offer a centralized platform on which students may post their theses online, the supervisors may review and comment on the theses, the evaluators may review and mark the thesis and the administrators may manage the users and records of the scholars. Role-based access control is employed to provide the security, data integrity and confidentiality in the system.

The project would result in improved transparency, fewer errors, and timely evaluation by removing the manual process and implementing an automated web portal to remove unnecessary administrative workload. The Thesis Evaluation Web Portal is a part of a modern, paperless and efficient academic evaluation system that is appropriate in the modern learning institutions.

#### **2.2 Related Work**

The article by Gupta et al. (2016) created MovementSlicer, which is a visualization tool on the patterns of behavior and schedules of meetings based on the improved Gantt charts. This system takes advantage of visualization which is data-rich to help in the analysis and optimization of the scheduling processes to provide more insight on a complex situation of scheduling. Its innovation is more visualization oriented rather than on the actual scheduling algorithm improvement[1].

Dong et al. (1998) developed a software algorithm to schedule classes in the university and the systematic strategies of allocation were stressed to solve the computational problems of the scheduling processes. This initial literature formed the basis of algorithmic techniques to timetabling but was not scalable and adaptable to the current, dynamic academic contexts [2].

As Lukas et al. (2009) showed, the method of solving university timetable problems with the

help of GA was complemented by heuristic search. Their approach was successful in solving such problems as resource conflicts and overbooking. They have emphasized scalability of GA with large data set which would be appropriate with complex university settings. Nevertheless, the method needed a lot of computing power to perform the best results [3].

The idea of the Thesis Evaluation Web Portal is associated with the already designed academic management systems like Student Information Systems (SIS), Learning Management Systems (LMS), and Research Management systems applied in universities. Some institutions have used partial digital solutions to assist in thesis submission, supervisor feedback and evaluation processes in the past.

Certain universities have embraced Learning Management System such as Moodle or Google Classroom to receive thesis documents and provide feedback. Nevertheless, such systems are not specifically created to evaluate structured theses, role-based or official result management. Other institutions use submission systems through email or internal portals which are not transparent, have no efficient tracking and record management.

Using the methodology of research and academic projects, the idea of online submission and assessment of the thesis has been suggested to save the paperwork and enhance efficiency. These systems are normally oriented to the submission of digital documents, supervisor verification and evaluator grading. Nevertheless, a lot of the available solutions are either limited in their capabilities, institution-specific or they are not integrated to have all stakeholders (students, supervisors, evaluators and administrators) on a single platform.

Even with such efforts, a fully integrated, secure and user friendly web based thesis evaluation system is still required with full workflow automation, real time status tracking, role based access control and high data security. The proposed while also seeking to enhance the current methodologies, the Thesis Evaluation Web Portal will provide a more streamlined, open and scalable solution specifically in regards to academic thesis evaluation.

## Related Work – Thesis Evaluation Systems

### Online Thesis Submission System



- Upload theses online
- Automated plagiarism checks
- Submission tracking

### Learning Management System



- Course materials access
- Student progress tracking
- Communication tools

### Research Management System



- Research project database
- Resource management
- Data analysis tools

### Paperless Evaluation System



- Digital thesis evaluation
- Online grading
- Feedback forms

Figure 2.2.1: Related work

## **CHAPTER 3**

### **METHODOLOGY**

#### **3.1 Introduction**

The software development methodology refers to a procedural or a set of processes which are applied in the development and maintenance of software systems. This is a very important element of any project development. In this section, I will address how my proposed system on thesis evaluation web portal will develop.

The thesis management Web Portal is split into a series of functional modules so that there is smooth and efficient management of the thesis evaluation process.

##### **1. User Management Module**

- User registration and login
- Role (Student, Supervisor, Evaluator, Admin).
- Profile management
- Authentication and authorization.

##### **2. Student Module**

- Thesis topic submission
- Thesis document upload (PDF)
- View supervisor feedback
- Monitor thesis assessment progress.
- View final results

##### **3. Supervisor Module**

- Review thesis submissions
- Accept/ decline proposal of thesis.
- Give feedback and recommendations.
- Monitor student progress

##### **4. Evaluator Module**

- Access assigned theses
- Submit evaluation reports
- Assign marks or grades



- Provide feedback

## **5. Management Module Thesis Submission.**

- Online thesis submission
- Document control versions.
- Status monitoring (Submitted, Under Review, Evaluated)
- Thesis files should be stored securely.

## **6. Evaluation & Grading Module**

- Evaluation form submission
- Marks/grade calculation
- Result approval process
- Final result generation

## **7. Admin Module**

- Manage users and roles
- Delegate supervisors and evaluators
- Manage academic sessions
- Create reports and statistics

## **8. Notification Module**

- Dashboard or email notifications
- Student and faculty status updates
- Assessment and notification of outcomes

## **9. Reporting Module**

- Thesis submission reports
- Evaluation summary reports
- Student performance report

## **3.2 Requirement Specification**

Requirement specification gives a detailed description of the objectives the automatization system of thesis submission must satisfy including the functional as well as the non-functional ones.

### **3.2.1 Functional Requirements**

Functional requirements tell about the services which the system should provide and the way in which it should respond to different inputs and act under certain conditions.

#### **Register/Sign up**

Admin: This will require the user to give their name, phone number, and email ID.

Teachers: Teachers have to fill in their name, phone, email address and a password.

Students: The students need to give their complete name, session, email address and a secure password.

#### **Authentication**

Input: The users complete a certain information when registering.

Output: Assure of successful registration whereby a user is informed of the success by a unique user id through email.

Processing: The system checks given details. In case of errors, corrective message is shown.

#### **Login**

Input: Enter email and password.

Output: Allows the user to access system features based on their role.

### **Student Functional Requirements**

1. The system will enable the students to create and update their profiles.
2. The system will enable students to come up with thesis titles and proposals.
3. The system will enable the students to post thesis papers in PDF.
4. The system will enable the students to see thesis submission status.
5. The system will enable students to see the comments of supervisors and examiners.

### **Supervisor Functional Requirements**

1. The system will enable the supervisors to see delegated students.
2. The system will enable the supervisors to reject or accept thesis topics.
3. The system will enable the supervisors to comment and give directions.
4. The system will enable the supervisors to monitor the student progress.

### **Thesis Submission Functional Requirements**

1. Thesis proposals shall be submitted in the system.

2. The system will be in a position to submit final thesis on time.
3. The system will authorize uploaded file type and size.
4. The system will not allow late submissions except by approval of the system by the admin.

### **3.2.2 Non-Functional Requirements**

Non-functional requirements are those that define the attribute and limitations that the system has to meet with other than its fundamental functionalities.

#### **Usability Requirement**

The user interface must be user friendly and accessible to admins, teachers and students. Prompts must be provided to the users when they are involved in input processes.

#### **Availability Requirement**

The system must be 24/7 to serve the needs of the users and accommodate changes whenever necessary.

#### **Efficiency Requirement**

The system must manage large datasets.

#### **Accuracy**

The system should provide accurate and current status of thesis with real time update in case there are any changes.

#### **Performance Requirement**

It should take no more than 3-5 seconds to load pages and present information in the system.

#### **Error Handling**

Examples of errors that the system should deal with include inputting the wrong data, having conflicting unavailable resources.

Show the right error messages to instruct the users.

#### **Security Requirements**

Only authorized admins should be able to access.

The data that includes should be encrypted and kept safe.

### 3.3 Process Model

The process model is a structure through which an abstract perspective of the process of creating a software system is given out. There are different models that could be used to guide the development process and these include waterfall model, iterative model, spiral model, and incremental model. To automate the process of class routine, the following models are taken into consideration:

#### Waterfall Model

The Waterfall model is the linear and sequential development model. The phases should be done sequentially and therefore it is appropriate in projects that have clear requirements. This model is however very risky and is not flexible since once the process is initiated the changes cannot easily be introduced.

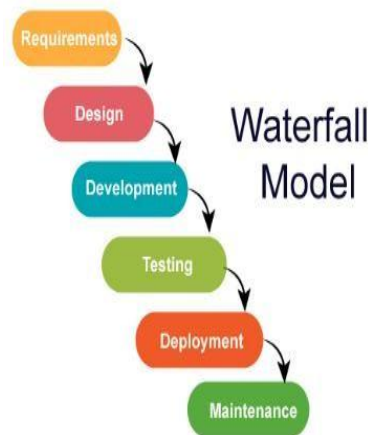


Figure 3.3.1: Waterfall Model

#### Iterative Model

The Iterative model is based on the principle of construction of a simple version of the system and the enhancement of it in several cycles. As it advances to new heights in implementation, the software becomes more functional and better. This is appropriate where requirements change or there is need in continuous inspirations



. Figure 3.3.2: Iterative Model

### Spiral Model

The Spiral model incorporates the waterfall model with the iterative development with focus on risk analysis. The spiral consists of phases of development, and each cycle (or loop) of the spiral involves planning, risk assessment and prototyping.

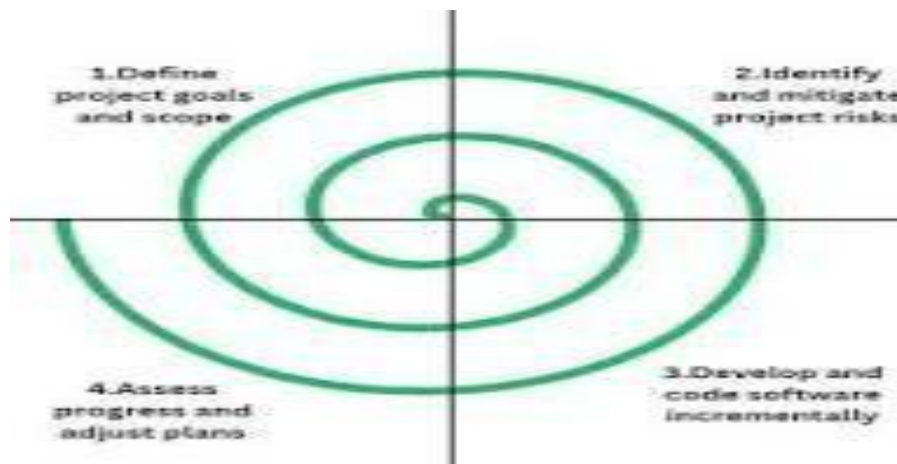


Figure 3.3.3: Spiral Model

### Incremental Process Model

The Incremental model divides the project into small manageable parts. Every component is created in a cyclic manner, and planning, development as well as testing takes place with each increment. This will make sure that when the system is being developed, a working system is provided early and then other features will be added with succeeding cycles.

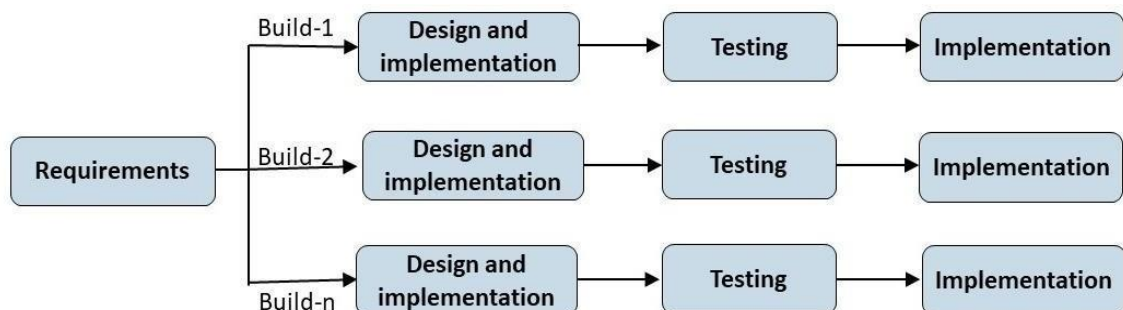


Figure 3.3.4: Incremental life cycle model

### **Reason for Choosing the Incremental Process Model:**

For thesis evaluation, the incremental model is the most suitable because:

1. The system requirements are well-defined and can be modularized.
2. While minor details may evolve, the key requirements are stable.
3. There is a need to deliver the system quickly to facilitate routine management.
4. The project involves integrating new technologies that can be incrementally tested and refined.

## **3.4 System Design**

The design of the system is based on the problem analysis and the requirements analysis. Upon this, I will attempt to engineer this system in a manner that will provide users with quicker system execution.

The system design is made up of 3 sections:

- i. Use case diagram design
- ii. E-R diagram
- iii. Activity flow diagram

Upon these designs, we write our system as per our preferred language of coding. The system design is known as coding the system based on the design model. My seminar library management system would have 6 modules referred to as the admin, teacher, students, book transaction, book module, report. This system will involve a user database that identifies a user to either be an admin teacher/students and redirect to their respective module.

The following tools that I will use to design the Use-case diagram, E-R diagram and Activity Diagram are as follows.

|                                 |                         |
|---------------------------------|-------------------------|
| Use-Case Diagram Designing Tool | EdrawMax                |
| E-R Diagram Designing Tool      | PlantUML                |
| Activity Diagram Tool           | Libeoffice Presentation |

Table 3.4.1: Tools used for Use-Case Diagram, E-R diagram and Activity diagram Design

### 3.4.1 Use-Case Diagram Design

Use case diagram shows how the users use the system and which part of the system they can have access to and can work with.

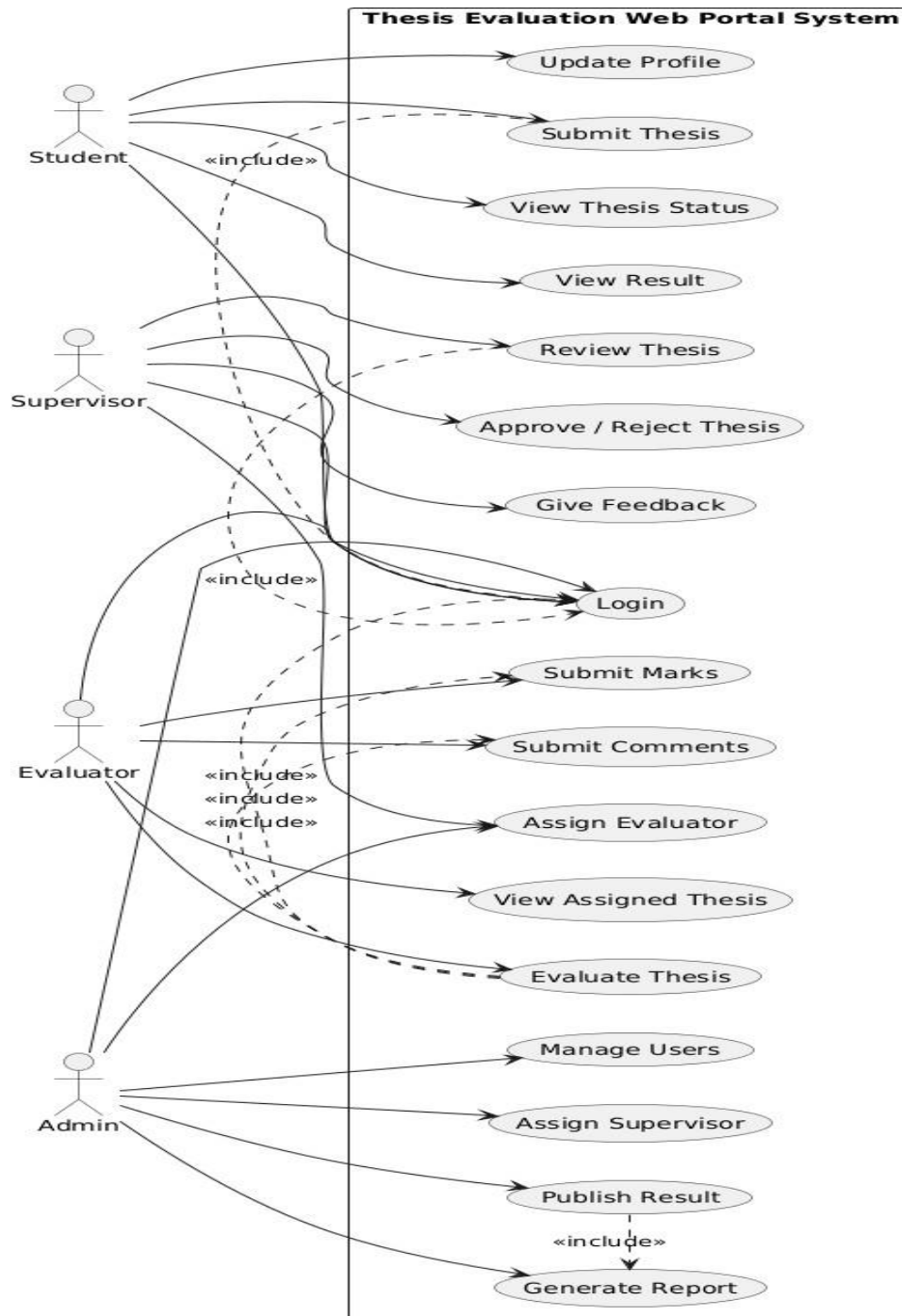


Figure 3.4.1.1: Use case diagram for thesis evaluation web portal

### 3.4.1 E-R Diagram Design

The E-R diagram shows how entity and their corresponding attributes are interrelated of each other. The entity-relation of the database model is shown by the following E-R diagram:

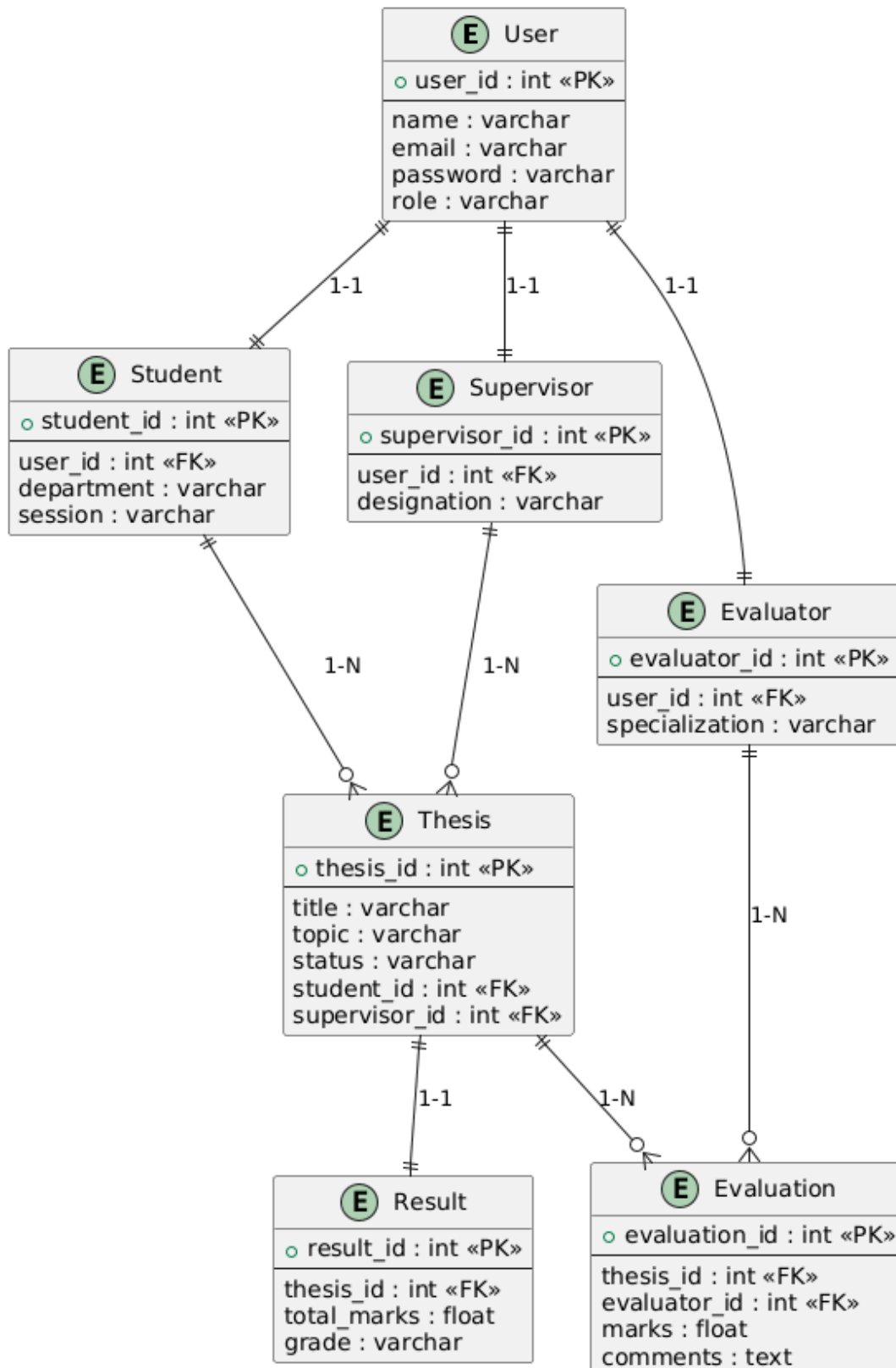


Figure 3.4.2.1: E-R diagram for thesis evaluation web portal



### 3.4.2 Activity Diagram Design

The flow from one step to the another one is represented by the activity diagram. An operation of the system could be used to characterized the activity. From one step to the next one, the flow is drawn. This flow can be sequential, branching or progress in the same time. Different elements like as fork, join and others are used in activity diagram to represent the direction of the flow.

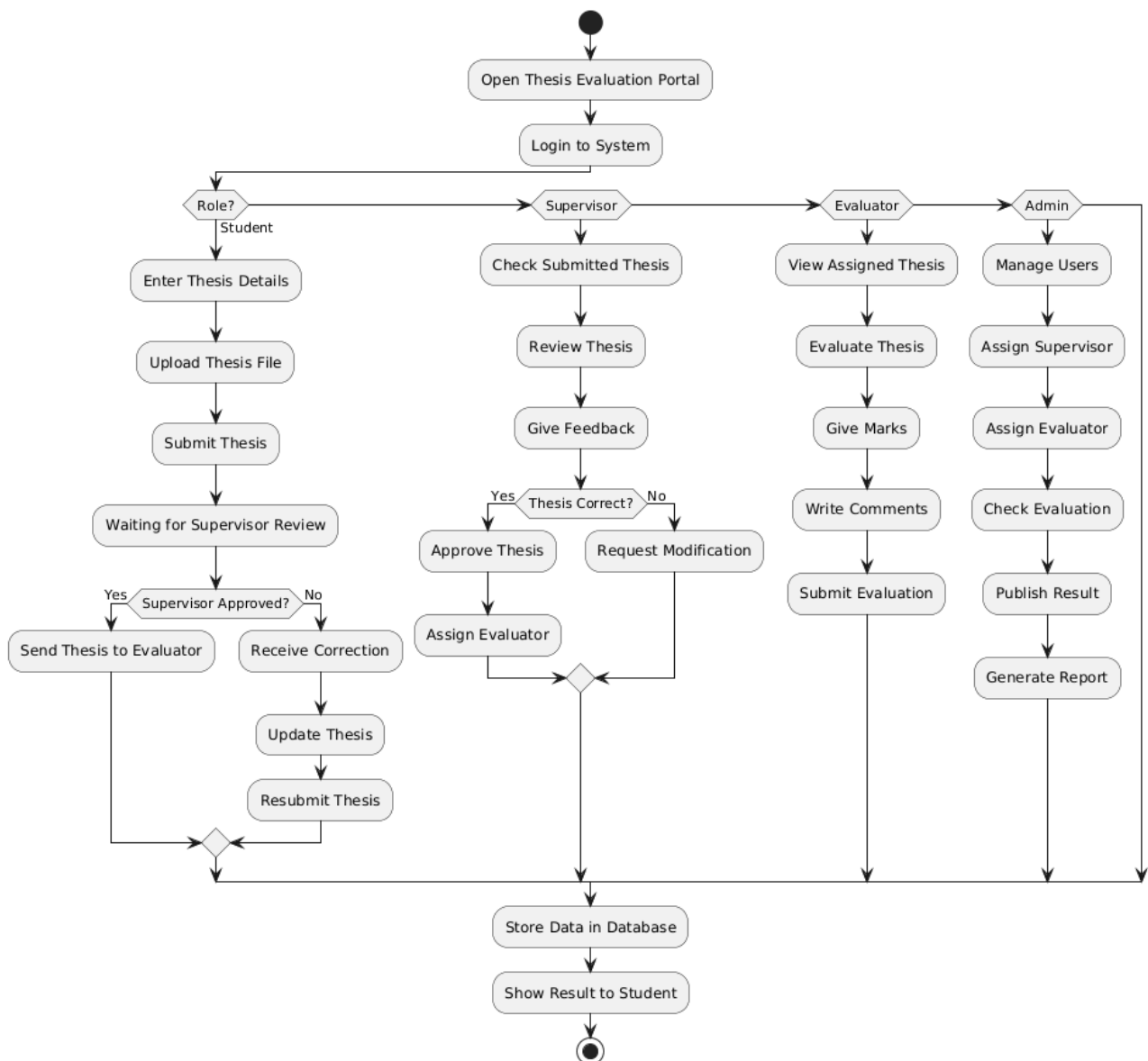


Figure 3.4.3.1: Activity Diagram for User and Admin Login

### 3.5 System Implementation

This process involves determining the appearance of the information system, ensuring that the system works well, and that it is quality. It includes the process of the system construction, making it work and taking care of its good quality.

**3.5.1 Software Requirements:** The following software will be used to complete the project:

|                  |                                                                 |
|------------------|-----------------------------------------------------------------|
| Operating system | Linux, Windows XP or later                                      |
| Browser          | Google Chrome, Firefox                                          |
| Editor           | Notepad++, Visual Studio Code, Subline Text or any text editor. |

Table 3.5.1.1: Software Requirement of the system

### 3.5.2 Hardware Requirements:

The following hardware will be used to complete the project

|           |                    |
|-----------|--------------------|
| Processor | Standard Processor |
| RAM       | 2GB RAM or more    |
| Hard Disk | 50 GB or more      |
| Monitor   | Standard monitor   |
| Keyboard  | Standard keyboard  |
| Mouse     | Standard mouse     |

Table 3.5.2.1: Hardware Requirements of the System

### 3.6 Technologies and Tools

In essence, this project will be done using MERN stack. This client-server architecture has two types of tools that can be used to develop this project. In order to come up with this software project, I will use the following technologies.

#### **Technologies for Client-Side Scripting:**

- HTML (Hypertext Markup Language)
- CSS (Cascading Style Sheets)
- Bootstrap (CSS Framework)
- JavaScript
- React JS (JavaScript Framework Library)

**Technologies Server-Side Scripting:**

- Node.js (JavaScript runtime environment)
- Express.js (JavaScript based back-end framework for Node.js)

**Hosting and version control system:**

- Heroku (for free hosting)
- GIT (for version control system)

**Database Related Tools:**

- MongoDB (NoSQL Database)
- Mongoose (MongoDB ODM)
- PHP(My SQL)

- (i) HTML: HTML or Hypertext Markup Language is the basic components of the World Wide Web. Its major use is to establish the significance and the organization of online material. CSS, on the contrary, is targeted at providing appearance to the web pages, whereas JavaScript improves their functionality and behavior. The Hypertext is used to define the interrelations linking web pages, which are inside of one web site, or they are between several sites. Such links are an inseparable part of the internet. By posting on the internet and connecting it to the pages that have been made by other users, you are already playing an active role in the immense universe of the World Wide Web. [7].
- (ii) CSS: CSS (Cascading Style Sheets) is a straightforward technique of enhancing the aesthetics of web documents e.g. fonts, colors and spacing [16]. The stylesheet language is applied to describe the appearance of HTML or XML documents in specifying the appearance of the elements in various mediums, including a screen, print, speech, and other media [8].
- (iii) Bootstrap: Bootstrap is an open source CSS framework to build responsive mobile-first front-end web application. It offers CSS-based design templates on the typography, forms, buttons, navigation as well as other interface elements and the ability to incorporate JavaScript [10].
- (iv) JavaScript: JavaScript is a computer programming language which is used to make websites more interactive. It can work with the device of the user and the web server. It converts a non-interactive web pages into the dynamic web pages, and improves the user experiences. JavaScript is widely used in web applications and browsers, but may also be used in software, server, and hardware controls. The user is able to manipulate web pages so as to do things like clicking. buttons, color changes, showing pictures, timers, animations, and dropdown menus.

Simply, HTML and CSS determine the frame and the look of a webpage whereas JavaScript provides functions that interact with the user. [11].

- (v) React: React is an open-source JavaScript library, which can be utilized to develop user interfaces (UI) and UI components. It is a flexible essential tool that is declared and component-based and makes it a flexible tool to develop single-page applications. React is a powerful JavaScript library that was initially launched by Facebook in the year 2013. It is also proficient in dealing with the interface layer and facilitating web and mobile development [12].
- (vi) Node.js: Node.js is an open-source, cross-platform, JavaScript runtime environment that runs JavaScript programs without the use of a web browser [13].
- (vii) Express.js: The express is a light-weight and flexible web application framework written in Node.js which gives a complete collection of tools to develop web and mobile applications with ease. It speeds up the creation of web applications written in Node.js adding a number of significant features.
- (v) MongoDB and Mongoose: MongoDB was also released as an open-source, cross-platform and document-oriented database program which is cross-platform. It is considered as the leading database solution to the contemporary applications. It is a no SQL database which operates on Jersey-like records with the liberty of optional designs. MongoDB is an open-source database that was developed by MongoDB Inc. and it is under the Server-Side Public License (SSPL) [23]. Mongoose is an Object Data Modeling (ODM) library created in the MongoDB ecosystem using Node.js. It also works like an Object Relational Mapper (ORM) such as SQL Alchemy, but is specific to MongoDB requirements. The main idea behind the Mongoose is to provide developers with the option to establish certain schemas at the application level. Other than schema enforcement, Mongoose contains numerous features, including hooks, model validation, and others [15].

### **3.7 Testing and Maintenance**

The main aim of testing is to find out errors. Testing implies trying to find out all of the possible mistakes or mistakes in what we have produced. It is like making sure that the software works as it is meant to work and breaks in a bad manner. Software testing involves validation and verification. Software testing has two approaches.

**i. Functionality testing or Black Box testing:** In this type of testing, the testers are not provided with any clue of how the software is developed internally, they are just required to assure that the software functions well on the surface. All depends on the way the software acts and works.

**ii. Implementation testing / White Box testing:** The White box testing (also referred to as structural testing) is applied to test the inner workings, circulation and structure of the software. In this testing, the tester develops test cases that are used to test the programs routes and logic flows in order to prove them to be as required or not[25].

### **3.7.1 Test objectives**

- i. All field data must be functioning well.
- ii. The links are supposed to take one to the required pages.
- iii. Moreover, one should not have interruptions in the input screen, messages or opinions.

### **3.7.2 Features to be tested**

- i. Make sure that every entry is made in a proper format.
- ii. Prevent repeated entry.
- iii. Make sure that all the links lead to the right pages.

### 3.8 Project Development Timetable

Here, I have given a timetable to complete my project. This is not an accurate timetable. It is an approximate timetable. I try my best to complete my project at this time.

| Activity Plan                            | Oct<br>25 | Nov<br>25 | Dec<br>25 | Jan<br>26 | Feb<br>26 | Mar<br>26 | Apr<br>26 | May<br>26 | Jun<br>26 | Jul<br>26 | Aug<br>26 |
|------------------------------------------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|
| Section of Project and Literature Review |           |           |           |           |           |           |           |           |           |           |           |
| User Interface Design                    |           |           |           |           |           |           |           |           |           |           |           |
| Front-end Development                    |           |           |           |           |           |           |           |           |           |           |           |
| Back-end Development                     |           |           |           |           |           |           |           |           |           |           |           |
| Testing and Modification s               |           |           |           |           |           |           |           |           |           |           |           |
| Documentation                            |           |           |           |           |           |           |           |           |           |           |           |

Table 3.8.1: Project Development Timetable

## **CHAPTER 4**

### **EXPECTED RESULT**

#### **4.1 Expected Result**

The expected result of the Thesis Evaluation Web Portal project is the successful development of a secure, efficient and user-friendly web-based system that automates the complete thesis evaluation process in an academic institution. The system is expected to replace traditional manual methods with a centralized digital platform.

After implementation, students will be able to submit their thesis online, track evaluation progress, and view feedback and final results without delays. Supervisors and evaluators will be able to review theses, provide comments and submit evaluations efficiently through role-based access. Administrators will be able to manage users, assign evaluators and generate reports accurately.

The portal is expected to:

- Reduce paperwork and manual workload
- Improve transparency and accountability in thesis evaluation
- Minimize human errors and data loss
- Ensure secure storage and management of thesis records
- Speed up the evaluation and result publication process

Overall, the Thesis Evaluation Web Portal is expected to enhance efficiency, accuracy and reliability in managing thesis evaluation activities and support a modern, paperless academic evaluation system.

## HOME PAGE,DASHBOARD

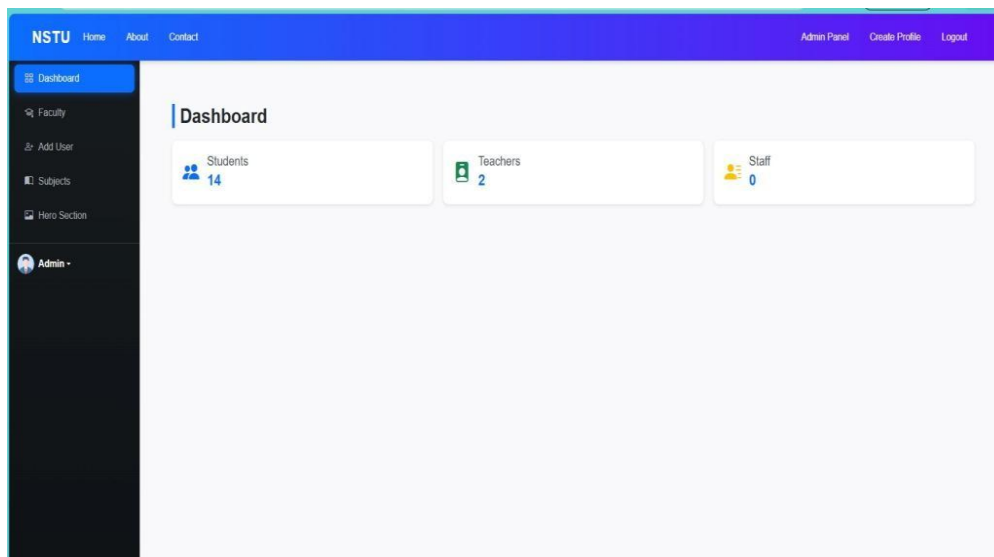
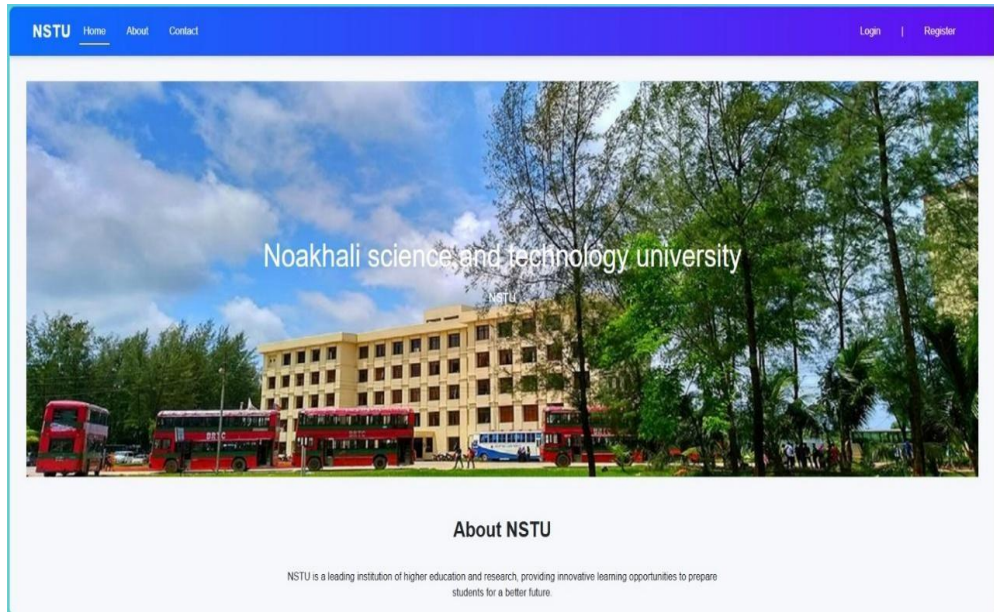
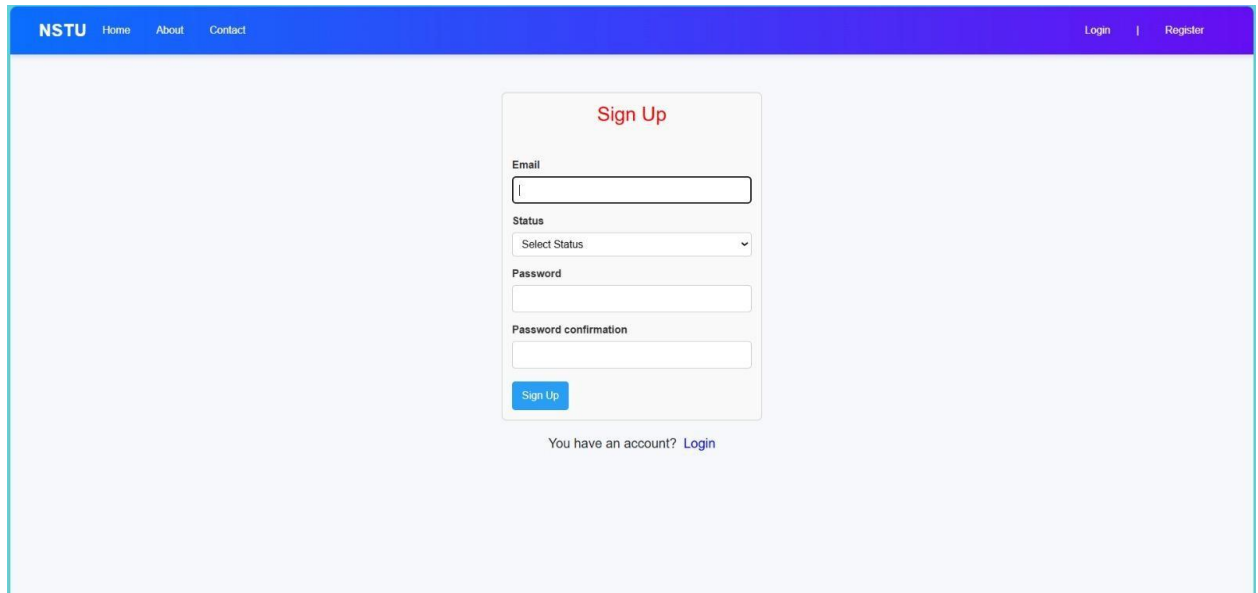


Figure 4.1.1: Home page,Admin Dashboard



## SIGN UP&LOG IN



The image shows a web page for signing up. At the top, there is a blue navigation bar with the text "NSTU" and links for "Home", "About", and "Contact" on the left, and "Login" and "Register" on the right. The main content area is light gray and contains a white "Sign Up" form. The form has a title "Sign Up" in red. It includes input fields for "Email", "Status" (a dropdown menu with "Select Status" and a downward arrow), "Password", and "Password confirmation". A blue "Sign Up" button is at the bottom of the form. Below the form, there is a link that says "You have an account? Login".

**Sign Up**

Email

Status  
Select Status ▼

Password

Password confirmation

[Sign Up](#)

You have an account? [Login](#)

Figure 4.1.2: Sign up page

Want to Sign up?

## Teacher Login Form

Email address

Password

Login

Forgot your password? [Reset](#)

No account yet? [Sign up](#)

Figure 4.1.3: Teacher login form

## **CHAPTER 5**

### **CONCLUSION AND FUTURE WORK**

#### **5.1 Conclusion**

The Thesis Management Web Portal provides an efficient, secure, and user-friendly platform for managing the entire thesis lifecycle in an academic environment. The system successfully automates key processes such as user management, thesis progress tracking, communication between students and supervisors, document handling and evaluation support, which are traditionally time-consuming and paper-based.

By integrating features like real-time progress monitoring, internal messaging, research paper recommendation and summarization tools, the portal enhances collaboration and improves the quality of thesis development. Students can manage their research work systematically while supervisors can easily monitor progress, provide timely feedback and maintain proper records.

The web-based nature of the system ensures accessibility, transparency, and centralized data management. Role-based access control increases security and ensures that each user interacts only with relevant functionalities. Overall, the Thesis Management Web Portal reduces administrative workload, minimizes errors and improves efficiency in thesis supervision and evaluation.

#### **5.2 Future Work**

Although the Thesis Management Web Portal successfully automates thesis monitoring, communication, and research support, there are several areas where the system can be enhanced in the future to make it more robust, intelligent and scalable.

One important extension is the integration of a plagiarism detection system to automatically verify the originality of thesis documents. This would help maintain academic integrity and reduce manual checking efforts. Additionally, AI-based evaluation support can be introduced to assist supervisors and examiners by analyzing writing quality, structure and relevance.

The system can be further improved by adding advanced analytics and dashboards that provide insights into student progress trends, supervisor workload and thesis completion rates. Such analytics would help institutions in decision-making and performance monitoring.

Another significant future enhancement is the development of a mobile application to allow students and supervisors to track progress, receive notifications and communicate anytime and anywhere. Integration with external academic databases such as Google Scholar, IEEE or Scopus could also improve the accuracy of research paper recommendations.

Security and scalability can be strengthened by implementing cloud-based storage, multi-factor

authentication, and blockchain-based document verification to ensure tamper-proof records. Furthermore, the portal can be expanded to support multi-university or multi-department systems, making it suitable for large academic institutions.

In the future, the system may also support digital signatures, automated approval workflows, and continuous feedback mechanisms to further streamline the thesis management process.

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